



Threat Hunting

1. Introduction

Threat hunting is a proactive cybersecurity practice that focuses on identifying hidden threats that bypass traditional detection systems. Threat hunting actively searches for indicators of compromise (IOCs), suspicious behaviors, and attacker tactics within an environment.

This report demonstrates both theoretical understanding and practical implementation of structured threat hunting methodologies using Elastic Security, Velociraptor, and threat intelligence sources.

The hunting activity aligns with techniques defined in the **MITRE Corporation ATT&CK Framework**, specifically technique **T1078 – Valid Accounts**, which involves misuse of legitimate credentials for unauthorized access.

2. Theoretical Knowledge

2.1 Proactive vs Reactive Security

Reactive Incident Response

- Triggered after alerts are generated.
- Responds to confirmed incidents.
- Focused on containment and remediation.

Proactive Threat Hunting

- Begins without alerts.
- Based on hypothesis-driven investigation.
- Aims to detect stealthy adversaries.

Threat hunting assumes that an attacker may already be present in the environment and actively searches for abnormal patterns.

2.2 Hypothesis-Driven Hunting

Hypothesis-driven hunting follows a structured approach:



1. Develop a security hypothesis.
2. Identify relevant data sources.
3. Execute targeted queries.
4. Validate suspicious findings.
5. Document results and refine detection rules.

Example Hypothesis:

“An attacker has misused valid domain credentials to escalate privileges.”

This maps directly to **MITRE ATT&CK T1078 – Valid Accounts**.

2.3 Threat Hunting Frameworks

SqRR Framework (Search, Query, Retrieve, Respond)

- **Search:** Identify relevant logs and telemetry.
- **Query:** Execute structured log queries.
- **Retrieve:** Collect suspicious artifacts.
- **Respond:** Mitigate or escalate findings.

TaHiTI Framework (Targeted Hunting Integrating Threat Intelligence)

- Integrates external threat intelligence.
- Correlates IOCs with internal logs.
- Enhances detection accuracy.

Reference methodologies are widely discussed in research publications from the **SANS Institute**.

2.4 Data Sources for Threat Hunting

Effective hunting relies on diverse telemetry sources:

- Windows Security Logs
- Event ID logs
- EDR data
- Network traffic
- Threat Intelligence feeds
- Endpoint process monitoring
- Authentication logs

For this exercise:



- Elastic Security was used for log querying.
- Velociraptor was used for endpoint investigation.
- AlienVault OTX was used for IOC validation.

3. Practical Implementation

3.1 Hunting Hypothesis

Hypothesis:

Unauthorized privilege escalation occurred using valid domain credentials.

The objective was to detect misuse of administrative privileges by analyzing Windows Event ID logs.

3.2 Log Query – Elastic Security

Event ID 4672 was selected for analysis.

Event ID 4672 indicates:

Special privileges assigned to new logon.

This is critical when investigating privilege escalation.

Query Executed:

Search filter for:

- Event ID = 4672
- Domain accounts
- Abnormal login times

3.3 Hunting Results

Timestamp	User	Event ID	Notes
2025-08-18 15:00:00	testuser	4672	Unexpected admin role assigned
2025-08-18 15:05:00	backupadm	4672	Normal scheduled access

Observation:

User **testuser** received elevated privileges outside normal operational hours. This behavior deviated from standard baseline activity.



3.4 Threat Intelligence Correlation

A suspicious source IP was identified during log review.

Steps performed:

1. Searched IP reputation using AlienVault OTX.
2. Checked related IOCs.
3. Cross-referenced endpoint processes using Velociraptor.

Velociraptor Query:

```
SELECT * FROM processes
```

Suspicious PowerShell activity was observed on the affected endpoint.

This strengthened the likelihood of credential misuse.

4. Analysis

The hunting investigation revealed:

- Unexpected administrative role assignment.
- Privilege escalation outside normal working hours.
- Suspicious PowerShell process execution.
- External IP flagged in threat intelligence feeds.

These findings indicate potential misuse of valid credentials, consistent with **MITRE ATT&CK T1078 – Valid Accounts**.

Although no full compromise was confirmed, the behavior was considered high risk and escalated for further monitoring.

5. Threat Hunting Report Summary

A proactive threat hunting exercise was conducted to identify potential misuse of valid domain credentials. Using Elastic Security, Event ID 4672 logs were analyzed to detect abnormal privilege assignments. An unexpected elevation of privileges for user “testuser” was identified outside regular operating hours. Further investigation using Velociraptor revealed suspicious PowerShell activity. The source IP was checked against AlienVault OTX



and showed malicious indicators. The activity was mapped to MITRE ATT&CK technique T1078 (Valid Accounts). While no confirmed breach occurred, the findings indicate potential credential misuse and require continuous monitoring and improved access control enforcement.

6. Key Learnings

- Threat hunting requires structured methodology.
 - Hypothesis-driven investigations improve accuracy.
 - Combining internal logs with external threat intelligence enhances detection.
 - Privilege escalation monitoring is critical.
 - Documentation strengthens SOC maturity.
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7. Recommendations

- Implement stricter privilege access controls.
 - Monitor Event ID 4672 continuously.
 - Enforce multi-factor authentication (MFA).
 - Conduct periodic threat hunting exercises.
 - Improve log retention and correlation capabilities.
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8. Conclusion

This exercise successfully demonstrated proactive threat hunting using structured frameworks and security tools. By applying hypothesis-driven analysis and correlating logs with threat intelligence, suspicious privilege escalation activity was identified and documented. Continuous threat hunting strengthens SOC readiness and reduces attacker dwell time within enterprise environments.
